Final Database Project

English Premier League 2027-2028 Season Database

Finley Hardie

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1. Business Description

This database is designed for a made-up English Premier League (EPL) statistics tracking system for the 2027-2028 season."PremierStats Analytics," is a sports analytics company that tracks individual player performance statistics across all 20 EPL clubs. The database serves scouts, coaches, analysts, and media professionals who need to access and compare player performance data across different positions and teams.

The prem is the topflight of football anywhere in the world. Consisting of 20 teams that each play 38 matches per season (home and away against every other team). Each team fields players across five positions: goalkeeper, defenders, midfielders, wingers, and strikers. Because the statistical metrics vary significantly by position (for example, save percentage is relevant only to goalkeepers, while successful take-ons apply to wingers), the database organizes player statistics by positional role rather than combining all players into a single table.

The database stores team-level information (standings, points, wins, draws, losses, goals for and against) as well as individual player season statistics appropriate to each position. With this setup, PremierStats Analytics can compare players within their positional groups, link individual stats to team results, and surface top performers across the league.

2. Business Rules

The following business rules govern the design and constraints of this database:

Rule 1: Each team in the league is uniquely identified by its TeamName. No two teams can share the same name.

Rule 2: Each player is uniquely identified by their PlayerName within their position table. A player can only appear in one positional table (a player is classified as either a goalkeeper, defender, midfielder, winger, or striker).

Rule 3: Every player must be associated with exactly one team. A team can have zero, one, or many players in each positional category. This creates a one-to-many `relationship from TEAM to each position table.

Rule 4: The TeamName column in each position table is a foreign key referencing the TEAM table. If a team name is updated, the change cascades to all associated player records (on update cascade). If a team is deleted, all player records for that team are also deleted (on delete cascade).

Rule 5: All statistics fields are required (not null) because every player who appears in the database must have a complete statistical record for the season.

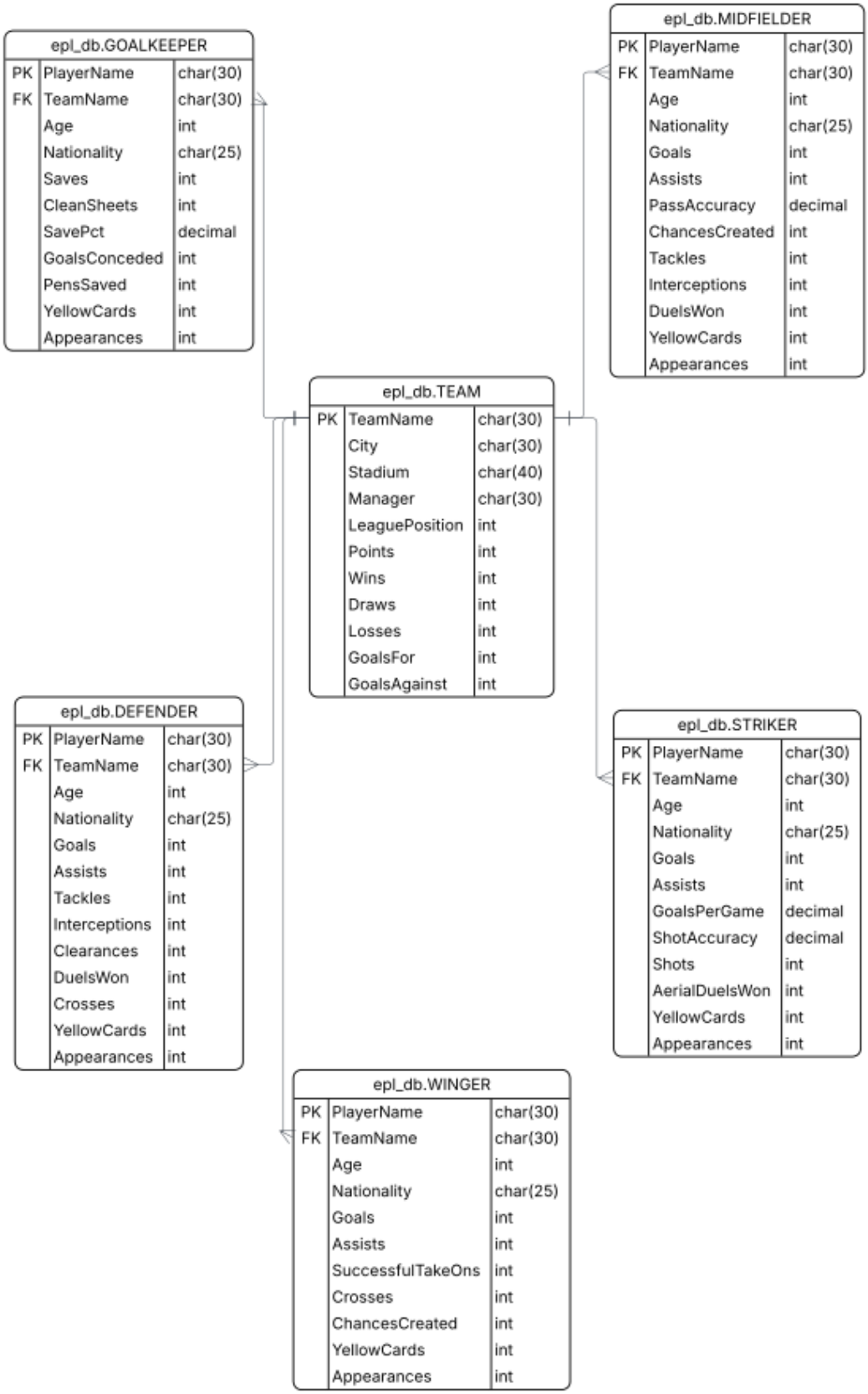
Rule 6: The LeaguePosition in the TEAM table must be a unique value between 1 and 20, representing the final standings of each team in the league.

Rule 7: Percentage-based statistics (SavePct, PassAccuracy, ShotAccuracy) are stored as decimal values representing the percentage (82.14 = 82.14%).

Rule 8: Each team plays 38 games per season. Player appearances can range from 0 to 38 depending on fitness, selection, and transfers.

3. ER Diagram

The following ER diagram shows the six entities in the database and the one-to-many relationships from the TEAM entity to each position entity. Each position table references the TEAM table through the TeamName foreign key. Primary keys are underlined, and foreign keys are labeled as FK.


epl_db.GOALKEEPER
PK PlayerName char(30)
FK TeamName char(30)
Age int
Nationality char(25)
Saves int
CleanSheets int
SavePct decimal
GoalsConceded int
PensSaved int
YellowCards int
Appearances int
epl_db.MIDFIELDER
PK PlayerName char(30)
FK TeamName char(30)
Age int
Nationality char(25)
Goals int
Assists int
PassAccuracy decimal
ChancesCreated int
Tackles int
Interceptions int
DuelsWon int
YellowCards int
Appearances int
epl_db.TEAM
PK TeamName char(30)
City char(30)
Stadium char(40)
Manager char(30)
LeaguePosition int
Points int
Wins int
Draws int
Losses int
GoalsFor int
GoalsAgainst int
epl_db.DEFENDER
PK PlayerName char(30)
FK TeamName char(30)
Age int
Nationality char(25)
Goals int
Assists int
Tackles int
Interceptions int
Clearances int
DuelsWon int
Crosses int
YellowCards int
Appearances int
epl_db.STRIKER
PK PlayerName char(30)
FK TeamName char(30)
Age int
Nationality char(25)
Goals int
Assists int
GoalsPerGame decimal
ShotAccuracy decimal
Shots int
AerialDuelsWon int
YellowCards int
Appearances int
epl_db.WINGER
PK PlayerName char(30)
FK TeamName char(30)
Age int
Nationality char(25)
Goals int
Assists int
SuccessfulTakeOns int
Crosses int
ChancesCreated int
YellowCards int
Appearances int

Cardinality Explanation: The TEAM entity has a 1:N relationship with each of the five position entities (GOALKEEPER, DEFENDER, MIDFIELDER, WINGER, STRIKER). This means one team can have many players in each position, but each player belongs to exactly one team. The foreign key TeamName in each position table enforces this relationship.

4. Functional Dependencies and Normalization

Functional Dependencies

TEAM Table:

```
TeamName → (City, Stadium, Manager, LeaguePosition, Points, Wins, Draws,
Losses, GoalsFor, GoalsAgainst)
```

TeamName is the only determinant. All other attributes depend on TeamName alone.

GOALKEEPER Table:

```
PlayerName → (TeamName, Age, Nationality, Saves, CleanSheets, SavePct,
GoalsConceded, PensSaved, YellowCards, Appearances)
```

PlayerName is the sole determinant. Every goalkeeper attribute depends on the individual player, not the team.

DEFENDER Table:

```
PlayerName → (TeamName, Age, Nationality, Goals, Assists, Tackles,
Interceptions, Clearances, DuelsWon, Crosses, YellowCards, Appearances)
```

PlayerName determines all other attributes. No defensive stat depends on anything other than the player.

MIDFIELDER Table:

```
PlayerName → (TeamName, Age, Nationality, Goals, Assists, PassAccuracy,
ChancesCreated, Tackles, Interceptions, DuelsWon, YellowCards, Appearances)
```

PlayerName is the only determinant. All midfield stats are tied to the individual player.

WINGER Table:

```
PlayerName → (TeamName, Age, Nationality, Goals, Assists, SuccessfulTakeOns,
Crosses, ChancesCreated, YellowCards, Appearances)
```

PlayerName is the sole determinant here. Winger stats follow the player, not the team.

STRIKER Table:

```
PlayerName → (TeamName, Age, Nationality, Goals, Assists, GoalsPerGame,
ShotAccuracy, Shots, AerialDuelsWon, YellowCards, Appearances)
```

PlayerName determines everything else. All striker statistics belong to the individual player.

Confirmation of 3NF

3NF: All tables are in Third Normal Form because there are no transitive dependencies. In each table, every non-key attribute depends directly on the primary key and on nothing else. For example, in the GOALKEEPER table, Saves does not determine CleanSheets (a goalkeeper can have many saves but few clean sheets, or vice versa). Similarly, in the TEAM table, Points does not determine LeaguePosition in a strict functional sense because goal difference can break ties. All non-key attributes are independent of each other and depend only on the primary key.

5. SQL to Create and Populate Tables

The following SQL statements create the six tables and populate them with fictional data for the 2027-2028 EPL season. All data uses real player names with fictional statistics.

```
-- 2027-2028 English Premier League SQLite Database
-- Man United dominates the season


-- CREATE TEAM TABLE
CREATE TABLE TEAM (
    TeamName CHAR(30) NOT NULL PRIMARY KEY,
    City CHAR(30) NOT NULL,
    Stadium CHAR(40) NOT NULL,
    Manager CHAR(30) NOT NULL,
    LeaguePosition INT NOT NULL,
    Points INT NOT NULL,
    Wins INT NOT NULL,
    Draws INT NOT NULL,
    Losses INT NOT NULL,
    GoalsFor INT NOT NULL,
    GoalsAgainst INT NOT NULL
);


-- CREATE GOALKEEPER TABLE
CREATE TABLE GOALKEEPER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Saves INT NOT NULL,
    CleanSheets INT NOT NULL,
    SavePct DECIMAL(5,2) NOT NULL,
    GoalsConceded INT NOT NULL,
    PensSaved INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

-- CREATE DEFENDER TABLE
CREATE TABLE DEFENDER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    Tackles INT NOT NULL,
    Interceptions INT NOT NULL,
    Clearances INT NOT NULL,
    DuelsWon INT NOT NULL,
    Crosses INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

-- CREATE MIDFIELDER TABLE
CREATE TABLE MIDFIELDER (
```

```
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    PassAccuracy DECIMAL(5,2) NOT NULL,
    ChancesCreated INT NOT NULL,
    Tackles INT NOT NULL,
    Interceptions INT NOT NULL,
    DuelsWon INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

-- CREATE WINGER TABLE
CREATE TABLE WINGER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    SuccessfulTakeOns INT NOT NULL,
    Crosses INT NOT NULL,
    ChancesCreated INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

-- CREATE STRIKER TABLE
CREATE TABLE STRIKER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    GoalsPerGame DECIMAL(4,2) NOT NULL,
    ShotAccuracy DECIMAL(5,2) NOT NULL,
    Shots INT NOT NULL,
    AerialDuelsWon INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

-- INSERT TEAM DATA (All 20 EPL teams)
INSERT INTO TEAM VALUES
('Manchester United', 'Manchester', 'Old Trafford', 'Erik ten Hag', 1, 95, 29, 8, 5, 98, 31),
('Manchester City', 'Manchester', 'Etihad Stadium', 'Pep Guardiola', 2, 87, 26, 9, 7, 89, 42),
('Arsenal', 'London', 'Emirates Stadium', 'Mikel Arteta', 3, 84, 25, 9, 8, 87, 44),
('Liverpool', 'Liverpool', 'Anfield', 'Arne Slot', 4, 82, 24, 10, 8, 85, 46),
('Chelsea', 'London', 'Stamford Bridge', 'Mauricio Pochettino', 5, 76, 22, 10, 10, 79, 52),
('Tottenham', 'London', 'Tottenham Hotspur Stadium', 'Ange Postecoglou', 6, 73, 21, 10, 11, 77, 55),
('Brighton', 'Brighton', 'Amex Stadium', 'Roberto De Zerbi', 7, 71, 20, 11, 11, 75, 58),
('Aston Villa', 'Birmingham', 'Villa Park', 'Unai Emery', 8, 68, 19, 11, 12, 72, 61),
('West Ham', 'London', 'London Stadium', 'David Moyes', 9, 65, 18, 11, 13, 68, 64),
('Fulham', 'London', 'Craven Cottage', 'Marco Silva', 10, 62, 17, 11, 14, 65, 67),
('Brentford', 'London', 'Gtech Community Stadium', 'Thomas Frank', 11, 59, 16, 11, 15, 62, 70),
('Everton', 'Liverpool', 'Goodison Park', 'Sean Dyche', 12, 56, 15, 11, 16, 59, 73),
```

```
('Nottingham Forest', 'Nottingham', 'City Ground', 'Nuno Espirito Santo', 13, 53, 14, 11, 17, 56, 76),
('Leicester City', 'Leicester', 'King Power Stadium', 'Steve Cooper', 14, 50, 13, 11, 18, 53, 79),
('Crystal Palace', 'London', 'Selhurst Park', 'Oliver Glasner', 15, 47, 12, 11, 19, 50, 82),
('Ipswich Town', 'Suffolk', 'Portman Road', 'Kieran McKenna', 16, 44, 11, 11, 20, 47, 85),
('Bournemouth', 'Bournemouth', 'Vitality Stadium', 'Gary ONeil', 17, 41, 10, 11, 21, 44, 88),
('Wolverhampton', 'Wolverhampton', 'Molineux Stadium', 'Gary ONeil', 18, 38, 9, 11, 22, 41, 91),
('Southampton', 'Southampton', 'St Marys Stadium', 'Russell Martin', 19, 26, 6, 8, 28, 35, 98),
('Luton Town', 'Luton', 'Kenilworth Road', 'Rob Edwards', 20, 23, 5, 8, 29, 32, 102);

-- INSERT GOALKEEPER DATA
INSERT INTO GOALKEEPER VALUES
('Andre Onana', 'Manchester United', 28, 'Cameroon', 145, 18, 82.14, 31, 3, 2, 38),
('Ederson', 'Manchester City', 30, 'Brazil', 138, 15, 80.23, 42, 2, 1, 38),
('David Raya', 'Arsenal', 28, 'Spain', 135, 16, 79.41, 44, 1, 2, 38),
('Alisson Ramses Becker', 'Liverpool', 31, 'Brazil', 132, 14, 78.57, 46, 2, 3, 38),
('Robert Sanchez', 'Chelsea', 27, 'Spain', 125, 12, 76.83, 52, 1, 4, 38),
('Alphonse Areola', 'Fulham', 31, 'France', 120, 11, 75.47, 67, 0, 5, 38),
('Lucas Bergvall', 'Tottenham', 26, 'Sweden', 118, 10, 74.68, 55, 1, 37, 38),
('Mads Hermansen', 'Leicester City', 24, 'Denmark', 110, 8, 72.55, 79, 0, 6, 38),
('Bart Verbruggen', 'Brighton', 22, 'Netherlands', 128, 13, 78.92, 58, 2, 2, 38),
('Emiliano Martinez', 'Aston Villa', 31, 'Argentina', 122, 12, 77.05, 61, 1, 3, 38),
('Lukasz Fabianski', 'West Ham', 39, 'Poland', 115, 9, 73.41, 64, 1, 4, 38),
('Kepa Arrizabalaga', 'Brentford', 29, 'Spain', 119, 10, 74.44, 70, 0, 5, 37);

-- INSERT DEFENDER DATA
INSERT INTO DEFENDER VALUES
('Aaron Wan-Bissaka', 'Manchester United', 26, 'England', 2, 4, 78, 42, 156, 68, 12, 3, 38),
('Lisandro Martinez', 'Manchester United', 25, 'Argentina', 3, 2, 82, 51, 189, 79, 8, 2, 37),
('Harry Maguire', 'Manchester United', 30, 'England', 4, 3, 76, 48, 172, 73, 6, 4, 38),
('Diogo Dalot', 'Manchester United', 24, 'Portugal', 2, 5, 81, 39, 145, 64, 15, 2, 36),
('John Stones', 'Manchester City', 30, 'England', 3, 2, 79, 44, 168, 71, 9, 3, 37),
('Ruben Dias', 'Manchester City', 27, 'Portugal', 2, 1, 80, 45, 175, 75, 7, 2, 38),
('Kyle Walker', 'Manchester City', 33, 'England', 1, 3, 77, 38, 142, 62, 11, 5, 35),
('Ben White', 'Arsenal', 26, 'England', 3, 2, 81, 42, 158, 69, 10, 2, 38),
('Gabriel Magalhaes', 'Arsenal', 26, 'Brazil', 4, 4, 79, 41, 172, 74, 8, 3, 38),
('Oleksandr Zinchenko', 'Arsenal', 27, 'Ukraine', 1, 3, 78, 35, 124, 58, 12, 4, 32),
('Virgil van Dijk', 'Liverpool', 32, 'Netherlands', 3, 5, 77, 38, 168, 76, 6, 3, 38),
('Andrew Robertson', 'Liverpool', 29, 'Scotland', 1, 4, 76, 36, 128, 55, 14, 4, 36),
('Trent Alexander-Arnold', 'Liverpool', 25, 'England', 2, 3, 78, 42, 152, 63, 16, 3, 37),
('Reece James', 'Chelsea', 24, 'England', 1, 2, 75, 40, 138, 61, 13, 5, 32),
('Marc Cucurella', 'Chelsea', 25, 'Spain', 2, 3, 76, 37, 131, 57, 11, 3, 35);

-- INSERT MIDFIELDER DATA
INSERT INTO MIDFIELDER VALUES
('Bruno Fernandes', 'Manchester United', 29, 'Portugal', 12, 16, 88.45, 68, 52, 28, 82, 4, 38),
('Scott McTominay', 'Manchester United', 27, 'Scotland', 8, 5, 81.23, 31, 74, 42, 65, 5, 38),
('Casemiro', 'Manchester United', 31, 'Brazil', 4, 3, 83.67, 25, 92, 51, 58, 3, 37),
('Christian Eriksen', 'Manchester United', 32, 'Denmark', 5, 8, 82.11, 42, 38, 22, 45, 2, 32),
('Rodri', 'Manchester City', 28, 'Spain', 6, 2, 89.34, 28, 78, 45, 52, 2, 38),
('Phil Foden', 'Manchester City', 24, 'England', 9, 11, 85.67, 52, 41, 25, 68, 3, 37),
('Kevin De Bruyne', 'Manchester City', 32, 'Belgium', 7, 9, 87.89, 48, 35, 21, 42, 4, 28),
('Declan Rice', 'Arsenal', 25, 'England', 4, 3, 84.21, 22, 65, 38, 48, 3, 38),
('Martin Odegaard', 'Arsenal', 25, 'Norway', 8, 7, 86.45, 51, 31, 19, 35, 2, 38),
('James Maddison', 'Tottenham', 27, 'England', 6, 9, 82.33, 44, 28, 15, 32, 4, 35),
('Yves Bissouma', 'Tottenham', 28, 'Mali', 3, 2, 83.56, 18, 71, 42, 38, 5, 38),
('Alexis Mac Allister', 'Brighton', 24, 'Argentina', 5, 4, 84.67, 25, 42, 28, 42, 3, 37),
('Mateo Kovacic', 'Manchester City', 30, 'Croatia', 4, 3, 88.12, 24, 45, 32, 38, 1, 36),
('Moises Caicedo', 'Chelsea', 21, 'Ecuador', 2, 1, 79.45, 15, 58, 35, 31, 6, 32),
('Enzo Fernandez', 'Chelsea', 23, 'Argentina', 3, 2, 80.56, 18, 52, 28, 25, 4, 30);

-- INSERT WINGER DATA
INSERT INTO WINGER VALUES
('Marcus Rashford', 'Manchester United', 26, 'England', 18, 9, 72, 35, 48, 5, 38),
```

```
('Jadon Sancho', 'Manchester United', 24, 'England', 14, 8, 68, 41, 39, 4, 37),
('Alejandro Garnacho', 'Manchester United', 20, 'Argentina', 11, 7, 65, 28, 31, 3, 36),
('Bukayo Saka', 'Arsenal', 22, 'England', 13, 10, 69, 45, 38, 3, 38),
('Leandro Trossard', 'Arsenal', 29, 'Belgium', 10, 7, 66, 32, 28, 4, 35),
('Mohamed Salah', 'Liverpool', 31, 'Egypt', 16, 11, 71, 52, 42, 4, 37),
('Luis Diaz', 'Liverpool', 27, 'Colombia', 12, 8, 67, 38, 35, 3, 36),
('Son Heung-min', 'Tottenham', 31, 'South Korea', 14, 9, 68, 43, 36, 5, 37),
('Dejan Kulusevski', 'Tottenham', 24, 'Sweden', 10, 7, 66, 31, 30, 4, 36),
('Jack Grealish', 'Manchester City', 29, 'England', 7, 8, 78, 38, 35, 3, 34),
('Bernardo Silva', 'Manchester City', 29, 'Portugal', 8, 6, 83, 29, 31, 4, 36),
('Solly March', 'Brighton', 24, 'England', 7, 5, 64, 22, 24, 2, 34),
('Kaoru Mitoma', 'Brighton', 27, 'Japan', 9, 6, 74, 28, 30, 2, 35),
('Eberechi Eze', 'Crystal Palace', 25, 'England', 8, 4, 70, 26, 32, 3, 36);

-- INSERT STRIKER DATA
INSERT INTO STRIKER VALUES
('Cristiano Ronaldo', 'Manchester United', 39, 'Portugal', 28, 6, 0.74, 68.45, 45, 19, 2, 38),
('Erling Haaland', 'Manchester City', 24, 'Norway', 24, 5, 0.63, 64.23, 52, 18, 4, 38),
('Kai Havertz', 'Arsenal', 24, 'Germany', 19, 4, 0.50, 58.34, 42, 12, 3, 38),
('Darwin Nunez', 'Liverpool', 25, 'Uruguay', 18, 3, 0.47, 56.78, 48, 16, 5, 38),
('Nicolas Jackson', 'Chelsea', 22, 'Senegal', 15, 3, 0.39, 52.45, 38, 13, 4, 38),
('Richarlison', 'Tottenham', 26, 'Brazil', 12, 4, 0.35, 48.56, 34, 11, 5, 34),
('Evan Ferguson', 'Brighton', 21, 'Northern Ireland', 11, 2, 0.30, 44.23, 32, 9, 3, 36),
('Ollie Watkins', 'Aston Villa', 28, 'England', 16, 5, 0.42, 54.67, 42, 14, 4, 38),
('Danny Ings', 'West Ham', 31, 'England', 13, 2, 0.35, 48.34, 28, 8, 3, 37),
('Alexandre Lacazette', 'Arsenal', 37, 'France', 8, 1, 0.21, 38.45, 20, 6, 2, 32),
('Callum Wilson', 'Bournemouth', 32, 'England', 10, 2, 0.27, 42.56, 24, 7, 4, 36),
('Matej Vydra', 'Ipswich Town', 32, 'Czech Republic', 7, 1, 0.19, 35.67, 18, 5, 3, 36);
```

6. SQL Queries

Query 1: Aggregate Function

This query calculates the average goals scored by strikers on each team to show which clubs have the highest-scoring forward lines. It uses the COUNT, AVG, and SUM aggregate functions with GROUP BY.

```
-- QUERY 1: AGGREGATE FUNCTION - Average goals scored by strikers on each team
-- This query finds the average goals scored by strikers on each team, helping identify top-scoring
clubs
SELECT
    t.TeamName,
    t.LeaguePosition,
    COUNT(s.PlayerName) as StrikerCount,
    AVG(s.Goals) as AvgGoalsPerStriker,
    SUM(s.Goals) as TotalStrikerGoals
FROM TEAM t
LEFT JOIN STRIKER s ON t.TeamName = s.TeamName
GROUP BY t.TeamName, t.LeaguePosition
ORDER BY TotalStrikerGoals DESC;
```

Query 2: Join

This query joins strikers with their team info to show the top scorers on the five highest-ranked teams. Scouts looking at talent from the leading clubs would get the most use out of this.

```
-- QUERY 2: JOIN QUERY - Match player stats to team information
-- This query joins strikers with their team information to show elite performers on championship teams
SELECT
    s.PlayerName,
    s.TeamName,
    s.Goals,
    s.Assists,
    s.GoalsPerGame,
    t.LeaguePosition,
    t.Points
FROM STRIKER s
JOIN TEAM t ON s.TeamName = t.TeamName
WHERE t.LeaguePosition <= 5
ORDER BY s.Goals DESC;
```

Query 3: Nested Query

This query uses a subquery to find all defenders on the top-5 teams in the current standings. It’s a quick way to pull up the best defensive talent from the leading clubs.

```
-- QUERY 3: NESTED QUERY (SUBQUERY) - Find defenders in top teams
-- This query uses a subquery to find all defenders who play for teams currently in the top 5
SELECT
    PlayerName,
    TeamName,
    Age,
    Tackles,
    Interceptions,
    Clearances
FROM DEFENDER
```

```
WHERE TeamName IN (
    SELECT TeamName FROM TEAM WHERE LeaguePosition <= 5
)
ORDER BY TeamName, Tackles DESC;
```

Query 4: Derived Table

This query uses a derived table (subquery in the FROM clause) to find teams whose strikers average more than 12 goals per season, then joins back to the TEAM table to include the standings. It shows which clubs have productive striker groups.

```
-- QUERY 4: DERIVED TABLE QUERY - Teams with high-scoring strikers
-- This query uses a derived table to find teams whose strikers average more than 12 goals per season
SELECT
    t.TeamName,
    t.LeaguePosition,
    t.Points,
    striker_stats.AvgStrikerGoals,
    striker_stats.NumStrikers
FROM TEAM t
JOIN (
    SELECT
        TeamName,
        AVG(Goals) as AvgStrikerGoals,
        COUNT(*) as NumStrikers
    FROM STRIKER
    GROUP BY TeamName
    HAVING AVG(Goals) > 12
) striker_stats ON t.TeamName = striker_stats.TeamName
ORDER BY striker_stats.AvgStrikerGoals DESC;
```

7. Complete SQL Code

The following is the complete SQL code without comments

```
CREATE TABLE TEAM (
    TeamName CHAR(30) NOT NULL PRIMARY KEY,
    City CHAR(30) NOT NULL,
    Stadium CHAR(40) NOT NULL,
    Manager CHAR(30) NOT NULL,
    LeaguePosition INT NOT NULL,
    Points INT NOT NULL,
    Wins INT NOT NULL,
    Draws INT NOT NULL,
    Losses INT NOT NULL,
    GoalsFor INT NOT NULL,
    GoalsAgainst INT NOT NULL
);

CREATE TABLE GOALKEEPER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Saves INT NOT NULL,
    CleanSheets INT NOT NULL,
    SavePct DECIMAL(5,2) NOT NULL,
    GoalsConceded INT NOT NULL,
    PensSaved INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

CREATE TABLE DEFENDER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    Tackles INT NOT NULL,
    Interceptions INT NOT NULL,
    Clearances INT NOT NULL,
    DuelsWon INT NOT NULL,
    Crosses INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);

CREATE TABLE MIDFIELDER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    PassAccuracy DECIMAL(5,2) NOT NULL,
    ChancesCreated INT NOT NULL,
    Tackles INT NOT NULL,
    Interceptions INT NOT NULL,
    DuelsWon INT NOT NULL,
```

```
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);


CREATE TABLE WINGER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    SuccessfulTakeOns INT NOT NULL,
    Crosses INT NOT NULL,
    ChancesCreated INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);


CREATE TABLE STRIKER (
    PlayerName CHAR(30) NOT NULL PRIMARY KEY,
    TeamName CHAR(30) NOT NULL,
    Age INT NOT NULL,
    Nationality CHAR(25) NOT NULL,
    Goals INT NOT NULL,
    Assists INT NOT NULL,
    GoalsPerGame DECIMAL(4,2) NOT NULL,
    ShotAccuracy DECIMAL(5,2) NOT NULL,
    Shots INT NOT NULL,
    AerialDuelsWon INT NOT NULL,
    YellowCards INT NOT NULL,
    Appearances INT NOT NULL,
    FOREIGN KEY (TeamName) REFERENCES TEAM(TeamName) ON UPDATE CASCADE ON DELETE CASCADE
);


INSERT INTO TEAM VALUES
('Manchester United', 'Manchester', 'Old Trafford', 'Erik ten Hag', 1, 95, 29, 8, 5, 98, 31),
('Manchester City', 'Manchester', 'Etihad Stadium', 'Pep Guardiola', 2, 87, 26, 9, 7, 89, 42),
('Arsenal', 'London', 'Emirates Stadium', 'Mikel Arteta', 3, 84, 25, 9, 8, 87, 44),
('Liverpool', 'Liverpool', 'Anfield', 'Arne Slot', 4, 82, 24, 10, 8, 85, 46),
('Chelsea', 'London', 'Stamford Bridge', 'Mauricio Pochettino', 5, 76, 22, 10, 10, 79, 52),
('Tottenham', 'London', 'Tottenham Hotspur Stadium', 'Ange Postecoglou', 6, 73, 21, 10, 11, 77, 55),
('Brighton', 'Brighton', 'Amex Stadium', 'Roberto De Zerbi', 7, 71, 20, 11, 11, 75, 58),
('Aston Villa', 'Birmingham', 'Villa Park', 'Unai Emery', 8, 68, 19, 11, 12, 72, 61),
('West Ham', 'London', 'London Stadium', 'David Moyes', 9, 65, 18, 11, 13, 68, 64),
('Fulham', 'London', 'Craven Cottage', 'Marco Silva', 10, 62, 17, 11, 14, 65, 67),
('Brentford', 'London', 'Gtech Community Stadium', 'Thomas Frank', 11, 59, 16, 11, 15, 62, 70),
('Everton', 'Liverpool', 'Goodison Park', 'Sean Dyche', 12, 56, 15, 11, 16, 59, 73),
('Nottingham Forest', 'Nottingham', 'City Ground', 'Nuno Espírito Santo', 13, 53, 14, 11, 17, 56, 76),
('Leicester City', 'Leicester', 'King Power Stadium', 'Steve Cooper', 14, 50, 13, 11, 18, 53, 79),
('Crystal Palace', 'London', 'Selhurst Park', 'Oliver Glasner', 15, 47, 12, 11, 19, 50, 82),
('Ipswich Town', 'Suffolk', 'Portman Road', 'Kieran McKenna', 16, 44, 11, 11, 20, 47, 85),
('Bournemouth', 'Bournemouth', 'Vitality Stadium', 'Gary ONeil', 17, 41, 10, 11, 21, 44, 88),
('Wolverhampton', 'Wolverhampton', 'Molineux Stadium', 'Gary ONeil', 18, 38, 9, 11, 22, 41, 91),
('Southampton', 'Southampton', 'St Marys Stadium', 'Russell Martin', 19, 26, 6, 8, 28, 35, 98),
('Luton Town', 'Luton', 'Kenilworth Road', 'Rob Edwards', 20, 23, 5, 8, 29, 32, 102);


INSERT INTO GOALKEEPER VALUES
('Andre Onana', 'Manchester United', 28, 'Cameroon', 145, 18, 82.14, 31, 3, 2, 38),
('Ederson', 'Manchester City', 30, 'Brazil', 138, 15, 80.23, 42, 2, 1, 38),
('David Raya', 'Arsenal', 28, 'Spain', 135, 16, 79.41, 44, 1, 2, 38),
('Alisson Ramses Becker', 'Liverpool', 31, 'Brazil', 132, 14, 78.57, 46, 2, 3, 38),
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('Robert Sanchez', 'Chelsea', 27, 'Spain', 125, 12, 76.83, 52, 1, 4, 38),
('Alphonse Areola', 'Fulham', 31, 'France', 120, 11, 75.47, 67, 0, 5, 38),
('Lucas Bergvall', 'Tottenham', 26, 'Sweden', 118, 10, 74.68, 55, 1, 37, 38),
('Mads Hermansen', 'Leicester City', 24, 'Denmark', 110, 8, 72.55, 79, 0, 6, 38),
('Bart Verbruggen', 'Brighton', 22, 'Netherlands', 128, 13, 78.92, 58, 2, 2, 38),
('Emiliano Martinez', 'Aston Villa', 31, 'Argentina', 122, 12, 77.05, 61, 1, 3, 38),
('Lukasz Fabianski', 'West Ham', 39, 'Poland', 115, 9, 73.41, 64, 1, 4, 38),
('Kepa Arrizabalaga', 'Brentford', 29, 'Spain', 119, 10, 74.44, 70, 0, 5, 37);

INSERT INTO DEFENDER VALUES
('Aaron Wan-Bissaka', 'Manchester United', 26, 'England', 2, 4, 78, 42, 156, 68, 12, 3, 38),
('Lisandro Martinez', 'Manchester United', 25, 'Argentina', 3, 2, 82, 51, 189, 79, 8, 2, 37),
('Harry Maguire', 'Manchester United', 30, 'England', 4, 3, 76, 48, 172, 73, 6, 4, 38),
('Diogo Dalot', 'Manchester United', 24, 'Portugal', 2, 5, 81, 39, 145, 64, 15, 2, 36),
('John Stones', 'Manchester City', 30, 'England', 3, 2, 79, 44, 168, 71, 9, 3, 37),
('Ruben Dias', 'Manchester City', 27, 'Portugal', 2, 1, 80, 45, 175, 75, 7, 2, 38),
('Kyle Walker', 'Manchester City', 33, 'England', 1, 3, 77, 38, 142, 62, 11, 5, 35),
('Ben White', 'Arsenal', 26, 'England', 3, 2, 81, 42, 158, 69, 10, 2, 38),
('Gabriel Magalhaes', 'Arsenal', 26, 'Brazil', 4, 4, 79, 41, 172, 74, 8, 3, 38),
('Oleksandr Zinchenko', 'Arsenal', 27, 'Ukraine', 1, 3, 78, 35, 124, 58, 12, 4, 32),
('Virgil van Dijk', 'Liverpool', 32, 'Netherlands', 3, 5, 77, 38, 168, 76, 6, 3, 38),
('Andrew Robertson', 'Liverpool', 29, 'Scotland', 1, 4, 76, 36, 128, 55, 14, 4, 36),
('Trent Alexander-Arnold', 'Liverpool', 25, 'England', 2, 3, 78, 42, 152, 63, 16, 3, 37),
('Reece James', 'Chelsea', 24, 'England', 1, 2, 75, 40, 138, 61, 13, 5, 32),
('Marc Cucurella', 'Chelsea', 25, 'Spain', 2, 3, 76, 37, 131, 57, 11, 3, 35);

INSERT INTO MIDFIELDER VALUES
('Bruno Fernandes', 'Manchester United', 29, 'Portugal', 12, 16, 88.45, 68, 52, 28, 82, 4, 38),
('Scott McTominay', 'Manchester United', 27, 'Scotland', 8, 5, 81.23, 31, 74, 42, 65, 5, 38),
('Casemiro', 'Manchester United', 31, 'Brazil', 4, 3, 83.67, 25, 92, 51, 58, 3, 37),
('Christian Eriksen', 'Manchester United', 32, 'Denmark', 5, 8, 82.11, 42, 38, 22, 45, 2, 32),
('Rodri', 'Manchester City', 28, 'Spain', 6, 2, 89.34, 28, 78, 45, 52, 2, 38),
('Phil Foden', 'Manchester City', 24, 'England', 9, 11, 85.67, 52, 41, 25, 68, 3, 37),
('Kevin De Bruyne', 'Manchester City', 32, 'Belgium', 7, 9, 87.89, 48, 35, 21, 42, 4, 28),
('Declan Rice', 'Arsenal', 25, 'England', 4, 3, 84.21, 22, 65, 38, 48, 3, 38),
('Martin Odegaard', 'Arsenal', 25, 'Norway', 8, 7, 86.45, 51, 31, 19, 35, 2, 38),
('James Maddison', 'Tottenham', 27, 'England', 6, 9, 82.33, 44, 28, 15, 32, 4, 35),
('Yves Bissouma', 'Tottenham', 28, 'Mali', 3, 2, 83.56, 18, 71, 42, 38, 5, 38),
('Alexis Mac Allister', 'Brighton', 24, 'Argentina', 5, 4, 84.67, 25, 42, 28, 42, 3, 37),
('Mateo Kovacic', 'Manchester City', 30, 'Croatia', 4, 3, 88.12, 24, 45, 32, 38, 1, 36),
('Moises Caicedo', 'Chelsea', 21, 'Ecuador', 2, 1, 79.45, 15, 58, 35, 31, 6, 32),
('Enzo Fernandez', 'Chelsea', 23, 'Argentina', 3, 2, 80.56, 18, 52, 28, 25, 4, 30);

INSERT INTO WINGER VALUES
('Marcus Rashford', 'Manchester United', 26, 'England', 18, 9, 72, 35, 48, 5, 38),
('Jadon Sancho', 'Manchester United', 24, 'England', 14, 8, 68, 41, 39, 4, 37),
('Alejandro Garnacho', 'Manchester United', 20, 'Argentina', 11, 7, 65, 28, 31, 3, 36),
('Bukayo Saka', 'Arsenal', 22, 'England', 13, 10, 69, 45, 38, 3, 38),
('Leandro Trossard', 'Arsenal', 29, 'Belgium', 10, 7, 66, 32, 28, 4, 35),
('Mohamed Salah', 'Liverpool', 31, 'Egypt', 16, 11, 71, 52, 42, 4, 37),
('Luis Diaz', 'Liverpool', 27, 'Colombia', 12, 8, 67, 38, 35, 3, 36),
('Son Heung-min', 'Tottenham', 31, 'South Korea', 14, 9, 68, 43, 36, 5, 37),
('Dejan Kulusevski', 'Tottenham', 24, 'Sweden', 10, 7, 66, 31, 30, 4, 36),
('Jack Grealish', 'Manchester City', 29, 'England', 7, 8, 78, 38, 35, 3, 34),
('Bernardo Silva', 'Manchester City', 29, 'Portugal', 8, 6, 83, 29, 31, 4, 36),
('Solly March', 'Brighton', 24, 'England', 7, 5, 64, 22, 24, 2, 34),
('Kaoru Mitoma', 'Brighton', 27, 'Japan', 9, 6, 74, 28, 30, 2, 35),
('Eberechi Eze', 'Crystal Palace', 25, 'England', 8, 4, 70, 26, 32, 3, 36);

INSERT INTO STRIKER VALUES
('Cristiano Ronaldo', 'Manchester United', 39, 'Portugal', 28, 6, 0.74, 68.45, 45, 19, 2, 38),
('Erling Haaland', 'Manchester City', 24, 'Norway', 24, 5, 0.63, 64.23, 52, 18, 4, 38),
('Kai Havertz', 'Arsenal', 24, 'Germany', 19, 4, 0.50, 58.34, 42, 12, 3, 38),
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('Darwin Nunez', 'Liverpool', 25, 'Uruguay', 18, 3, 0.47, 56.78, 48, 16, 5, 38),
('Nicolas Jackson', 'Chelsea', 22, 'Senegal', 15, 3, 0.39, 52.45, 38, 13, 4, 38),
('Richarlison', 'Tottenham', 26, 'Brazil', 12, 4, 0.35, 48.56, 34, 11, 5, 34),
('Evan Ferguson', 'Brighton', 21, 'Northern Ireland', 11, 2, 0.30, 44.23, 32, 9, 3, 36),
('Ollie Watkins', 'Aston Villa', 28, 'England', 16, 5, 0.42, 54.67, 42, 14, 4, 38),
('Danny Ings', 'West Ham', 31, 'England', 13, 2, 0.35, 48.34, 28, 8, 3, 37),
('Alexandre Lacazette', 'Arsenal', 37, 'France', 8, 1, 0.21, 38.45, 20, 6, 2, 32),
('Callum Wilson', 'Bournemouth', 32, 'England', 10, 2, 0.27, 42.56, 24, 7, 4, 36),
('Matej Vydra', 'Ipswich Town', 32, 'Czech Republic', 7, 1, 0.19, 35.67, 18, 5, 3, 36);


SELECT
    t.TeamName,
    t.LeaguePosition,
    COUNT(s.PlayerName) as StrikerCount,
    AVG(s.Goals) as AvgGoalsPerStriker,
    SUM(s.Goals) as TotalStrikerGoals
FROM TEAM t
LEFT JOIN STRIKER s ON t.TeamName = s.TeamName
GROUP BY t.TeamName, t.LeaguePosition
ORDER BY TotalStrikerGoals DESC;


SELECT
    s.PlayerName,
    s.TeamName,
    s.Goals,
    s.Assists,
    s.GoalsPerGame,
    t.LeaguePosition,
    t.Points
FROM STRIKER s
JOIN TEAM t ON s.TeamName = t.TeamName
WHERE t.LeaguePosition <= 5
ORDER BY s.Goals DESC;


SELECT
    PlayerName,
    TeamName,
    Age,
    Tackles,
    Interceptions,
    Clearances
FROM DEFENDER
WHERE TeamName IN (
    SELECT TeamName FROM TEAM WHERE LeaguePosition <= 5
)
ORDER BY TeamName, Tackles DESC;


SELECT
    t.TeamName,
    t.LeaguePosition,
    t.Points,
    striker_stats.AvgStrikerGoals,
    striker_stats.NumStrikers
FROM TEAM t
JOIN (
    SELECT
        TeamName,
        AVG(Goals) as AvgStrikerGoals,
        COUNT(*) as NumStrikers
    FROM STRIKER
    GROUP BY TeamName
    HAVING AVG(Goals) > 12
) striker_stats ON t.TeamName = striker_stats.TeamName
```

```
ORDER BY striker_stats.AvgStrikerGoals DESC;
```